

Assignment 5

Shapnil Das

June 10, 2026

1 Question:

Simulate 500 independent Random Walks. Regress the 1st random walk on the other 499 (one at a time). Create a histogram of the 500 resulting R^2 values. What percentage of these regressions yielded a t-statistic greater than 1.96? If these variables are totally unrelated, why isn't that percentage 5%?

2 Solution:

First, we will use loops to generate 500 Independent Random Walks for 500 different shocks. A Random Walk process can be defined as the sum of all current and past shocks. These shocks are independent and has a non-decaying effect on the series.

$$y_t = y_{t-1} + \varepsilon_t$$

$$y_t = \sum_{i=1}^t \varepsilon_i \quad \text{for } y_0 = 0.$$

where ε_t is a white noise error term.

we have generated 500 independent shock $\varepsilon_i \sim \mathcal{N}(0, 1)$ for $i = 1, 2, \dots, 500$. And for each shock we have generated 500 independent random walk process y_t for $t = 1, 2, \dots, 500$ as the sum of all current and past shocks for 100 initial sample or observations $T = 100$.

Now we regress the first random walk y_1 on the other 499 random walks one at a time and calculate the R^2 and $T - statistic$ values for each regression.

$$y_1 = \alpha + \beta y_j + u_t \quad \text{for } j = 2, 3, \dots, 500$$

Thus we will have 499 sample statistics which can mimic a Monte-Carlo simulation of the distribution of R^2 and $T - statistic$ values when the variables are totally unrelated.

Our assumption about the R^2 score is as the random walks are independent, theoretically there should be any relation between y_1 and y_j for $j = 2, 3, \dots, 500$. The expected value of R^2 should be close to zero and the expected value of the $T - statistics$ should be significantly close to zero 95% of the time. considering the rejection region is 1.96 is in absolute value at 5% significance level, we should expect to not reject the null hypothesis of no relationship between y_1 and y_j for only 5% of the regressions.

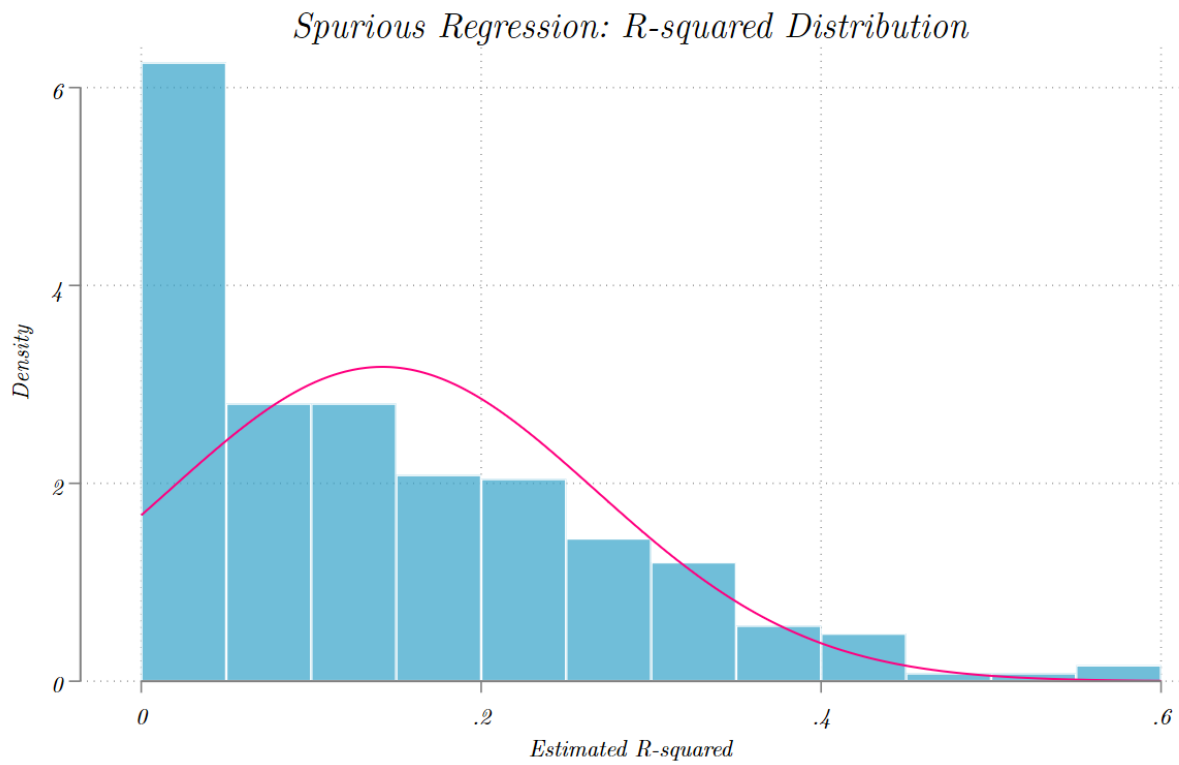
Table 1: Summary Statistics

Variable	Obs	Mean	Std. Dev.
Reject	499	0.731	0.444
R^2	499	0.142	0.126

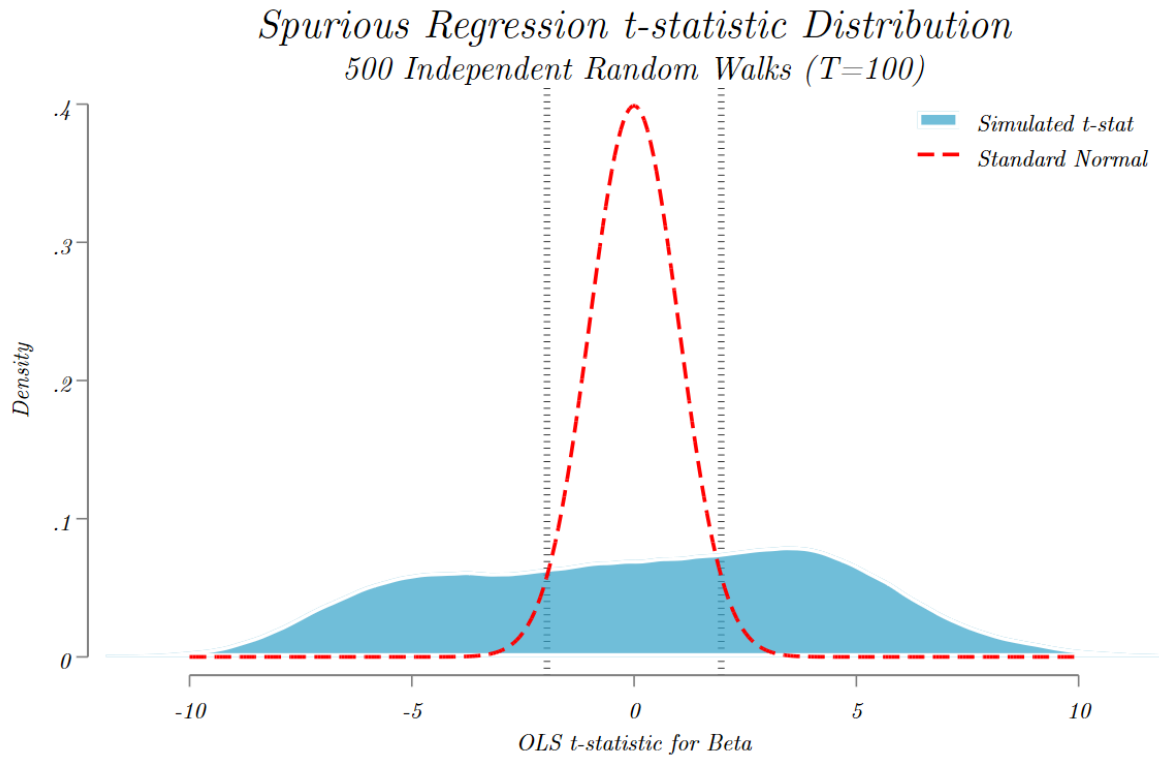
By Investigating the summary statistics we can see contradictions to our expectations. The T-statistics is rejecting the null hypothesis of no relationship between y_1 and y_j for 73.1% of the regressions which is much higher than the expected value in 5% significance level. And the R^2 values shows the independent random walks explains 14.2% of the variation in the first random walk.

This contradictory results are known as spurious regression.

A standard normal distribution of the shock contains a constant mean and constant variance but the shock of a random walk process follows a constant mean and a time dependent infinite variance where $\varepsilon_i \sim \mathcal{N}(\mu, t\sigma_i^2)$. Random walks have a non-decaying effect of all current and past shocks on the series, which can lead to create artificial relations with other unrelated random walks. A small shocks can change the trajectory of a random walk but it can never revert to it's original mean and wanders away to new trajectory randomly as time grows. By pure chance, 2 or more random walks can move in the same or opposite direction. So when we regress one on another, random walks, the regression can falsely identify the correlation among them as causation or relations. Thus they create an artificial goodness of fit r^2 and significant t-statistics which leads to spurious regression results. This is one of the core drawbacks of using random walk non-stationary process in Time Series analysis.



By plotting the histogram of the R^2 values we can see that most of the values are close to zero but there are some outliers which are far from zero and creates a long right skewed tail in the distribution. These outliers are the result of spurious regression which creates a significant t-statistics and high values of R^2 even when the variables are totally unrelated.

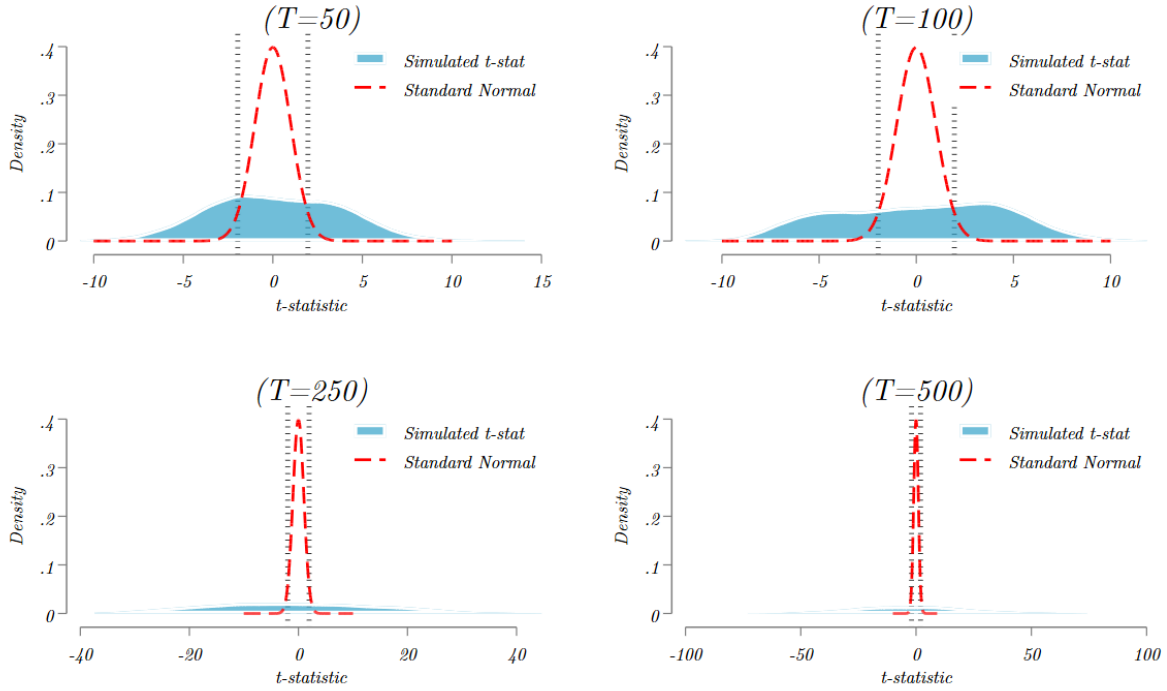


From the kernel density plot of the T-statistics we can see the contrast comparison of the standard normal distribution of the t-stat concentrated around zero and the distribution of the t-statistics from the regressions which is more spread out and presenting an uniform distribution, meaning standard significance tests will falsely tell us the variables are related. The 2 black dashed line represents the 2 tail rejection area at 5% significance level at ± 1.96 . For our empirical distribution of t-statistics, we can see that a large portion of the distribution falls outside the rejection area which is the result of spurious regression.

Table 2: Summary Statistics by Sample Size

Variable	Obs	Mean	Std. Dev.
T = 50			
Reject	499	0.653	0.476
R^2	499	0.183	0.170
T = 100			
Reject	499	0.731	0.444
R^2	499	0.142	0.126
T = 250			
Reject	499	0.936	0.245
R^2	499	0.373	0.261
T = 500			
Reject	499	0.956	0.205
R^2	499	0.389	0.277

Kdensity by sample size



Now to check by increasing the sample size, if we get any benefit with the spurious regression problem we have used 4 different sample size of 50, 100, 250, 500 and simulated each sample 500 times with a loop to regress first random walk on 499 others one at a time. we can see from the diagram and summary statistics that as we increase the sample size the problem increases. At lowest sample $T = 50$ the R^2 values are lower and the rejection percentage is 65%. we expect this values to lower as we increase the sample size to 500 but we can see at $T = 500$ the R^2 value is 38% and the rejection percentage is 95% at 5% significance level. Also from the diagram we can see that by increasing the sample size the distribution of the simulated values are more spread out are far away from the true normal distribution. We can conclude from our analysis that , for spurious regress increasing the sample will not solve the non-stationarity problem rather we have to difference the series to get proper reliable results.

```
1
2  cd "F:\ECON 408\Practice do
   file\Assignments\Time-Series-Assignment\Assugment-45\Do
   file"
3
4  clear all
5  set more off
6  set seed 10
7
8  drop _all
9  set obs 100
10 gen t = _n
11 tsset t
12
13 set scheme white_tableau
14 graph set window fontface "CMU Serif"
15 graph set eps fontface "CMU Serif"
16 graph set svg fontface "CMU Serif"
17
18
19 * Generating 500 Random walk process with a loop
20
21 forvalues i = 1/500 {
22     gen double u_`i' = rnormal()
23     gen double y_`i' = sum(u_`i')
24 }
25
26
27 * Creating columns and appending 400 rows to store 500
   regression results
28
29 gen t_stat = .
30 gen r2 = .
31 tsappend , add(400)
32
33 forvalues i = 2/500 {
34     quietly {
35         reg y_1 y_`i'
36         replace t_stat = _b[y_`i'] / _se[y_`i'] in `i'
37         replace r2 = e(r2) in `i'
38     }
39 }
```

```

40
41
42 * Test Statistics and R2 Squared analysis
43
44 gen reject = (abs(t_stat) > 1.96) if (t_stat !=.)
45
46 display "Target False Rejection Rate: 0.05 (5%)"
47 sum reject
48
49 display "Average R-squared of completely independent
variables:"
50 sum r2
51
52
53 * Visualizations
54
55 histogram r2, ///
56     width(0.05) ///
57     fcolor(ebblue%70) ///
58     lcolor(white) lwidth(vthin) ///
59     normal ///
60     normopts(lcolor(pink)) ///
61     title("Spurious Regression: R-squared Distribution",
color(black) size(medlarge)) ///
62     xtitle("Estimated R-squared") ///
63     ytitle("Density") name(histogram_hw5 , replace)
64
65 graph export histogram_hw5.png, name(histogram_hw5) replace
66
67
68 twoway ///
69     (kdensity t_stat, recast(area) fcolor(ebblue%70)
lcolor(white) lwidth(medthick)) ///
70     (function y = normalden(x), range(-10 10) lcolor(red)
lwidth(medthick) lpattern(dash)), ///
71     title("Spurious Regression t-statistic Distribution")
///
72     subtitle("500 Independent Random Walks (T=100)") ///
73     xtitle("OLS t-statistic for Beta") xlabel(,nogrid) ///
74     ytitle("Density") ylabel(,nogrid) ///
75     legend(order(1 "Simulated t-stat" 2 "Standard Normal")
col(1) pos(2) ring(0)) ///

```



```
76      xline(-1.96 1.96, lcolor(black) lpattern(dot) lwidth(  
      thick)) name(kdensity_hw5,replace)  
77  
78  graph export kdensity_hw5.png , name(kdensity_hw5) replace  
79  
80  
81  
82
```